# Data Augmentation in Prototypical Networks for Forest Tree Species Classification Using Airborne Hyperspectral Images

Long Chen, Yuxin Wei, Zongqi Yao, Erxue Chen, *Member, IEEE*, and Xiaoli Zhang, *Member, IEEE*

***Abstract*— Accurate and fine multiple tree species supervised classification based on few-shot learning has attracted close attention from researchers, because the sample collection is often hindered in forests. Prototypical networks (P-Nets), as a simple but efficient few-shot learning method, have significant advantages in forest tree species classification. Nevertheless, the overfitting phenomenon caused by the lack of training samples is still prevalent in few-shot classifiers, which brings challenges to training accurate classification models. In this study, we proposed a novel Proto-MaxUp (PM) framework to minimize the issue of overfitting from the perspective of data augmentation and a feature extraction backbone for tree species classification. Taking Gaofeng Forest Farm (GFF) in Nanning City, Guangxi Province, as the study area, nine tree species, cutting site, and road were classified. First, by analyzing the effects of a series of popular data augmentation methods and their combinations in different parts of the P-Net, several effective data augmentation pools were established. Then, the pools aforementioned were combined with PM to obtain the best classification performance. To verify the robustness and validity of the proposed strategy, we applied PM to the other four popular public hyperspectral datasets and achieved excellent results. Finally, this efficient data augmentation method was used in different feature extraction backbones. The results show that the classification accuracy was greatly improved with the optimal backbone (overall accuracy (OA) and Kappa, are 98.08% and 0.9789, respectively), and the difference between training accuracy and test accuracy is less than 2%. It is concluded that the accurate and fine classification for multiple tree species can be realized by the PM data augmentation strategy and backbone proposed in this article.**

***Index Terms*— Airborne hyperspectral images, data augmentation, MaxUp, prototypical networks (P-Nets), tree species classification.**

## I. Introduction

IDENTIFICATION of tree species using hyperspectral images (HSIs) has always been an important application in the domain of forestry. Airborne HSIs have the characteristics of high spatial resolution and continuous diagnostic spectral features, which can express rich spatial–spectral information [1]. However, in the forest with a complex stand structure, it is unlikely to identify vegetation types with very similar spectral reflectance of the same genus according to HSIs directly. The acquisition of the tree species classification samples is usually limited due to the complex terrain and inefficient field surveys, which makes it very difficult to label a large number of samples on the image and leads to the imbalance between the number of samples and the feature dimensions of the HSI. Few-shot learning refers to a task in which the classifier must be trained by decomposing the dataset into different meta-tasks (given only a few examples of each class) to adapt to the new tasks not seen in the training. Therefore, the few-shot learning is often proposed to obtain stronger model generalization when the number of samples is quite small. Prototypical networks (P-Nets), as a simple but effective few-shot learning method, have achieved encouraging results in tree species classification of HSIs in recent years [2], [3]. However, it is well known that the models trained by few-shot learning are particularly prone to overfitting [4], which brings challenges to training accurate tree species classification models. Therefore, in this article, we will explore the combination strategy of popular data augmentation methods to overcome the dilemma of overfitting in P-Net for tree species classification using the airborne HSIs.

Addressing the overfitting problem to improve model generalization potential is a central theme in the deep learning domain. Some regularization techniques introduce penalties for the complicacy of the model, which is usually equal to enforcing specific smoothing operations in the feature extraction part, such as Dropout [5], DropConnect [6], maxout [7], StochasticDepth [8], DropPath [9], ScheduledDropPath [10], shake–shake regularization [11], ShakeDrop regularization [12], and DropBlock [13]. As another mainstream form of regularization technology, data augmentation uses the important invariance properties of the data directly to improve performance [14]. In the domain of computer vision, data augmentation is almost ubiquitous due to its ease of implementation and effectiveness. Data augmentation was applied to deep learning, which first appeared in LeNet-5 for image affine transformation. With the development of convolutional neural network (CNN), various networks adopt data augmentations for image classification

Manuscript received January 30, 2022; revised March 12, 2022 and April 13, 2022; accepted April 13, 2022. Date of publication April 18, 2022; date of current version May 2, 2022. This work was supported in part by the National Key Research and Development Program of China, Ministry of Science and Technology, under Grant 2017YFD0600900; and in part by the Dragon 5 Cooperation under Grant 59257. *(Corresponding author: Xiaoli Zhang.)*

Long Chen, Yuxin Wei, Zongqi Yao, and Xiaoli Zhang are with the Precision Forestry Key Laboratory of Beijing and the Key Laboratory for Silviculture and Conservation of Ministry of Education, Beijing Forestry University, Beijing 100083, China (e-mail: charleychen@bjfu.edu.cn; WeiYuxin1119@bjfu.edu.cn; iceyaozq@bjfu.edu.cn; zhang-xl@263.net).

Erxue Chen is with the Institute of Forest Resource Information Techniques, Chinese Academy of Forestry, Beijing 100091, China (e-mail: chenerx@ifrit.ac.cn).

Digital Object Identifier 10.1109/TGRS.2022.3168054

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

tasks to prevent overfitting. For example, AlexNet used sample rescaling, random cropping, horizontal flipping, and color jittering to augment the training data [15], while the Visual Geometry Group (VGG) network used multiscaling and cropping [16]. GoogLeNet is the same as AlexNet in the training phase and also performs data augmentations to augment 1 image to 144 images during testing [17]. Similarly, in the ResNet [18] and DenseNet [19], these traditional methods based on geometric transformation were used to augment the image data, and significant accuracy improvement was also achieved. With the effectiveness of data augmentation verified in more and more work, many researchers began to study new data augmentation methods. In the field of gray image (handwritten characters) and natural image classification, many popular data augmentation methods, such as Synthetic Minority Over-sampling TEchnique (SMOTE) [20], SamplePairing [21], CutOut [22], Random Erasing [23], GridMask [24], MixUp [25], and CutMix [26], were proposed. Among them, SMOTE is mainly used to solve the problem of samples unbalanced between categories, which is applied to the intermediate features. The rest of methods can be divided into single data augmentation and multidata augmentation. CutOut, Random Erasing, and GridMask can only operate on a single sample to solve the problem of data occlusion, while SamplePairing, MixUp, and CutMix were applied to two or more images without reducing the proportion of informative pixels on training images [25]. In contrast, there are fewer data augmentation methods used in HSIs. For instance, Chen *et al.* [27] generated new virtual samples by adding noise to a sample and interpolating between two samples from the same category [27]. In the literature [28], a pixel pair strategy was used to generate a new sample with a new label by merging two samples from different categories. Other methods mostly generate new samples by adding noise [29], interpolation [29], [30], or using the relationship between sample points and neighborhoods [31]. These methods have been overlapped with the advanced data augmentations mentioned previously in the image classification field, such as adding noise with CutOut, interpolation with SamplePairing and MixUp, and neighborhood relations with CutMix.

In the classification task based on P-Net, however, the literature on data augmentation for HSI is fairly limited. The existing works on few-shot HSI classification use only rotation, vertical flipping, and horizontal flipping to augment images, which is parallel to the traditional augmentation methods [32], [33]. Although the effectiveness of conventional data augmentation methods has been verified in the few-shot HSI classification [2], [34]–[36], these methods are inadequate to cover the original feature distribution and learn new distribution, because the data augmentations are just for a single sample and the differences between samples are not considered. On the other hand, the works aforementioned do not decompose the classification task, which leads to the uncertainty of a suitable augmentation technique and its combination for each part of the task. The rapid development of newly P-Net-based networks is certainly beneficial to the generalization of the few-shot models [2], [37], [38]. We believe that the model of P-Net can achieve a significant performance improvement through introducing reasonable data augmentation strategies in each part of the networks. Consequently, the motivation for this study is to explore the data augmentation methods and their combination strategy suitable for each data input (e.g., support set, query set, and new tasks) of the P-Net and combine the most efficient feature extraction backbone currently available to minimize the issue of overfitting in the P-Net classifier, so as to improve the tree species classification accuracy with HSI under complex forest composition. Our work is summarized as follows.

1) According to the data inputs of the P-Net, we define three data augmentation modes (support augmentation, query augmentation, and inter-class augmentation) and discuss the optimal data augmentation methods of each mode.
2) Based on these experiments, we combine data augmentations that can obtain better performance than the baseline (no augmentation [NA]) to expand the data augmentation pool.
3) We introduce a MaxUp strategy in P-Net, named Proto-MaxUp (PM), and achieve further performance boosts for tree species classification using the airborne HSIs.
4) For P-Net with different backbones, we compare the classification performance of NA and using PM strategy to augment the training data, to define an optimal architecture for tree species classification.

## II. Materials

### A. Data Description

Five airborne hyperspectral datasets, Gaofeng Forest Farm (GFF), Indian Pines (IP), Kennedy Space Center (KSC), Botswana, and Chikusei, are used to experimentally verify the method we proposed. Due to the differences between these datasets in the distinguishability and the number of samples to be classified, for example, GFF is a dataset of multiple tree species, which has just subtle discrimination among each coniferous or broad-leaved tree species and the number of samples is fewer, while Chikusei is more discriminable than IP, KSC, and Botswana since its higher spatial resolution and larger sample size, we used a higher sample proportion for the GFF dataset and an extremely lower ratio of samples for the Chikusei dataset to train the model. The details are as follows.

*1) GFF:* The AISA Eagle II hyperspectral sensors were equipped by the Chinese Academy of Forestry's (CAF) LiCHy system to acquire HSIs over the sub-region of the Jiepai branch of GFF in Nanning City, Guangxi Province, China, on January 13 and 30, 2019 (Fig. 1). These data have 125 bands, which were sampled with wavelengths ranging from 400 to 987 nm. The high spatial–spectral resolution (1 m and 3.3 nm, respectively) aims to achieve fine-grained discrimination between tree species. The detailed parameters are referred to the literature [2], [3]. The dataset contains a variety of tree species; among them, coniferous species includes *Cunninghamia lanceolate*, *Pinus elliottii*, *Pinus massoniana*, and broadleaf species includes *Eucalyptus urophylla*, *Eucalyptus grandis*, *Castanopsis hystrix*, *Acacia melanoxylon*, *Mytilaria laosensis*, and other soft broadleaf species. Other

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

TABLE I
TRAINING AND TESTING SET DIVISION OF THE IP DATASET

| Class Name | Training | Testing |
|---|---|---|
| Alfalfa | 2 | 44 |
| Corn-notill | 71 | 1357 |
| Corn-mintill | 41 | 789 |
| Corn | 12 | 225 |
| Grass-pasture | 24 | 459 |
| Grass-trees | 37 | 693 |
| Grass-pasture-mowed | 1 | 27 |
| Hay-windrowed | 24 | 454 |
| Oats | 1 | 19 |
| Soybean-notill | 49 | 923 |
| Soybean-mintill | 123 | 2332 |
| Soybean-clean | 30 | 563 |
| Wheat | 10 | 195 |
| Woods | 63 | 1202 |
| Buildings-Grass-Trees-Drives | 19 | 367 |
| Stone-Steel-Towers | 5 | 88 |
| Total | 512 | 9737 |

TABLE II
TRAINING AND TESTING SET DIVISION OF THE KSC DATASET

| Class Name | Training | Testing |
|---|---|---|
| Scrub | 38 | 723 |
| Willow-swamp | 12 | 231 |
| CP-hammock | 13 | 243 |
| Slash-pine | 13 | 239 |
| Oak-Broadleaf | 8 | 153 |
| Hardwood | 11 | 218 |
| Swap | 5 | 100 |
| Graminoid-marsh | 22 | 409 |
| Spartina-marsh | 26 | 494 |
| Cattail-marsh | 20 | 384 |
| Salt-marsh | 21 | 398 |
| Mud-flats | 25 | 478 |
| Water | 46 | 881 |
| Total | 260 | 4951 |

land cover types include cutting site and road. In this way, each class contains 112 samples and 1232 samples in total. We divide the labeled samples into a training set and a testing set with a ratio of 2:8. It is worth noting that we only achieved about 92% testing accuracy of overall accuracy (OA) under the same sample proportion in our previous study [2].

*2) IP:* The AVIRIS sensor was used to acquire the IP dataset in Northwestern Indiana, USA, in 1992. This image contains 145 × 145 pixels with a spatial resolution of 20 m and 220 bands with wavelengths ranging from 400 to 2500 nm. In this article, we divide the training set and testing set according to the proportion of 5% and 95% in the ground truth (the background pixels were removed), as shown in Table I.

*3) KSC:* Similar to IP, the image of KSC was collected by the AVIRIS sensor over the KSC in Florida, USA, in 1996. In our experiments, the noisy bands (water absorption and low signal-to-noise ratio (SNR) bands) were removed. The KSC dataset has 512 × 614 pixels and 176 bands with wavelengths ranging from 400 to 2500 nm, and the spatial resolution is 18 m. We also take 5% labeled samples for the training set and 95% for the testing set. The quantity of the samples for 13 classes in the KSC dataset is shown in Table II.

*4) Botswana:* The Botswana dataset is an HSI set of various land cover types, acquired by the Hyperion sensor embedded on the NASA Earth Observing 1 satellite over the Okavango Delta, Botswana [40]. This image set has 1476 × 256 pixels and a pixel resolution of 30 m over a 7.7-km strip in 242 bands covering the 400–2500-nm portion of the spectrum in 10-nm windows. Uncalibrated and noisy bands that cover water absorption features were removed, and the remaining 145 bands were included as candidate features [41]. The samples in the dataset are divided into a training set and a testing set according to the proportion of 5% and 95%. Table III summarizes the distribution of the samples for 14 classes in the Botswana dataset.

TABLE III
TRAINING AND TESTING SET DIVISION OF THE BOTSWANA DATASET

| Class Name | Training | Testing |
|---|---|---|
| Water | 14 | 256 |
| Hippo Grass | 5 | 96 |
| Floodplain Grasses 1 | 13 | 238 |
| Floodplain Grasses 2 | 11 | 204 |
| Reeds | 13 | 256 |
| Riparian | 13 | 256 |
| Firescar | 13 | 246 |
| Island Interior | 10 | 193 |
| Acacia Woodlands | 16 | 298 |
| Acacia Shrublands | 12 | 236 |
| Acacia Grasslands | 15 | 290 |
| Short Mopane | 9 | 172 |
| Mixed Mopane | 13 | 255 |
| Exposed Soils | 5 | 90 |
| Total | 162 | 3086 |

TABLE IV
TRAINING AND TESTING SET DIVISION OF THE CHIKUSEI DATASET

| Class Name | Training | Testing |
|---|---|---|
| Water | 28 | 2817 |
| Bare soil (school) | 29 | 2830 |
| Bare soil (park) | 3 | 283 |
| Bare soil (farmland) | 48 | 4804 |
| Natural plants | 43 | 4254 |
| Weeds in farmland | 11 | 1097 |
| Forest | 205 | 20311 |
| Grass | 65 | 6450 |
| Rice field (grown) | 134 | 13235 |
| Rice field (first stage) | 13 | 1255 |
| Row crops | 60 | 5901 |
| Plastic house | 22 | 2171 |
| Manmade (non-dark) | 12 | 1208 |
| Manmade (dark) | 77 | 7587 |
| Manmade (blue) | 4 | 427 |
| Manmade (red) | 2 | 220 |
| Manmade grass | 10 | 1030 |
| Asphalt | 8 | 793 |
| Paved ground | 1 | 144 |
| Total | 775 | 76817 |

*5) Chikusei:* The Chikusei dataset comprises images of agricultural and urban areas in Chikusei, Ibaraki, Japan. The Headwall's Hyperspec-VNIR-C imaging sensor took the images on July 19, 2014 [42]. The dataset contains 2517 × 2335 pixels and 128 bands in the spectral range of 363–1018 nm with 1.29-nm spectral resolution and 2.5-m spatial resolution. The samples in the dataset are divided into a training set and a testing set according to the proportion of 1% and 99%. Table IV summarizes the distribution of the samples for 19 classes in the Chikusei dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

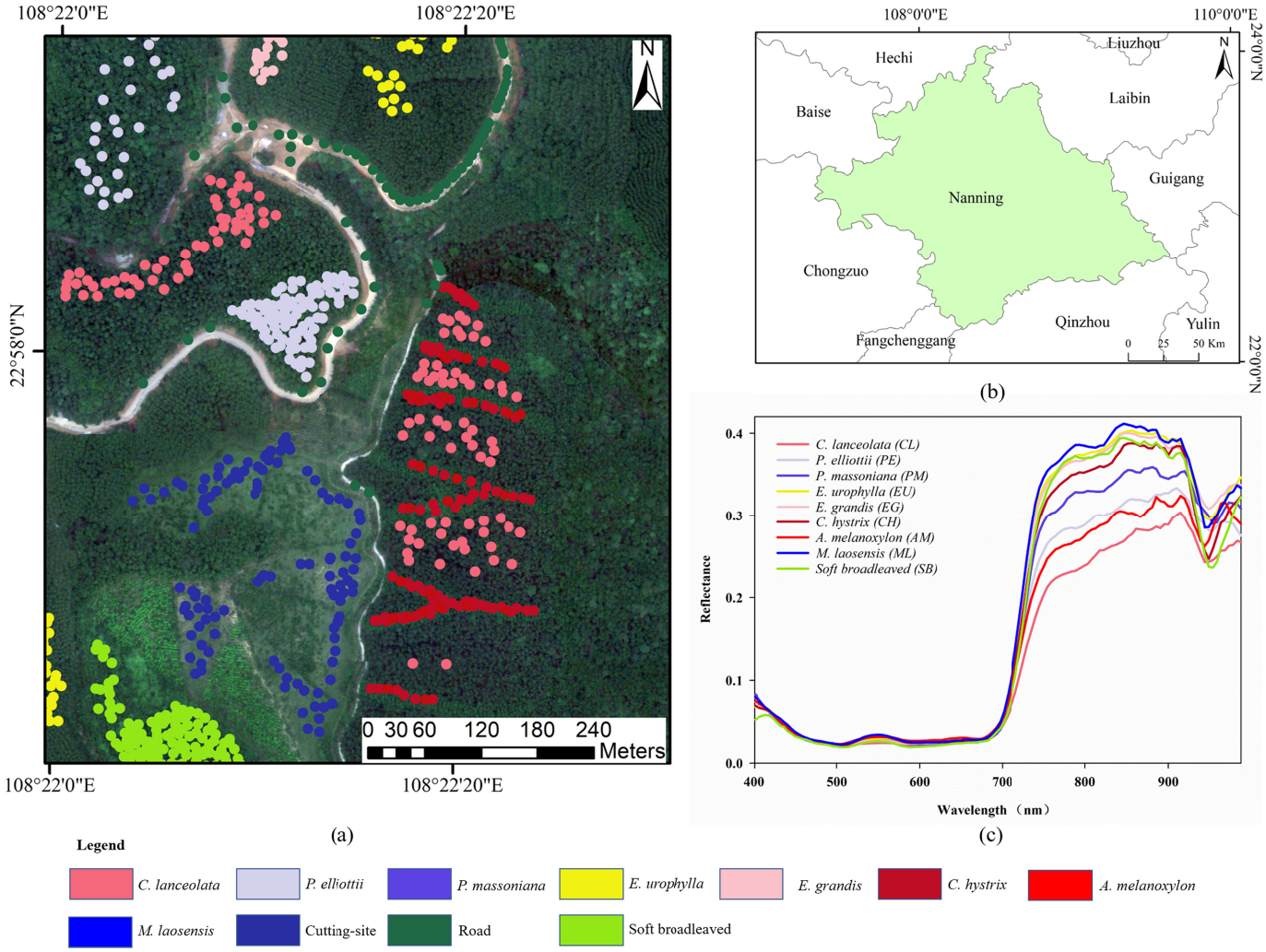



Fig. 1. Overview of the GFF dataset. (a) Combination image of hyperspectral data (red: band of 640.06 nm, green: band of 550.70 nm, and blue: band of 467.94 nm), and the samples distribution. (b) Location of Nanning City, Guangxi Province. (c) Average spectral reflectance curve of each tree species.

## III. Methodology

### A. Data Augmentation Modes in P-Net

As a simple and efficient few-shot classification method, P-Net aims to simulate multiple few-shot tasks through subsampling classes as well as data points. For an $N$-way classification task ($T_i$) in the P-Net, the support set ($S_i$) and query set ($Q_i$) are randomly sampled from the training dataset. Similar to most meta-learning methods, each iteration in the training process of the P-Net includes an internal loop and an external loop. The model first calculates the prototype and fine-tunes the parameters in the internal loop using the support set $S_i$. Then, the loss of parameters is minimized, and the prototype is optimized by evaluating the distance between the query set $Q_i$ and the corresponding prototype in the external loop [39]. Motivated by this training procedure, we can augment data in support set, query set, and the categories that are used in simulation tasks. In this article, we divided these data augmentations into three modes.

*1) Support Augmentation (Inner Class):* Data augmentations were applied to support set in the inner class. This mode expands the distribution range of prototypes in the internal loop, but does not change the diversity of training tasks.

*2) Query Augmentation (Inner Class):* When the query set in the inner class uses data augmentations, the prototype optimization process in the external loop becomes more sufficient. By this mode, the sampled evaluation data in the process of model training are expanded, so that the few-shot learner can learn a more comprehensive sample distribution than the training task without query augmentation.

*3) Inter-Class Augmentation:* We add a new class through some data augmentation methods that can operate between classes, such as SamplePairing, MixUp, and CutMix, to enrich the diversity of training tasks.

Fig. 2 shows the classification architecture of the P-Net with data augmentations, including data input and subsampling classes to tasks, data augmentations, image encoder, distance measurement, and classification. At the top of the architecture, we describe the process of data input and subsampling classes to tasks in a training task. The dataset is sampled into $N$ tasks ($\{T_1, \ldots, T_N\}$). Each task ($T_i$) contains a support set ($\{S_{i1}, \ldots, S_{in}\}$) and a query set ($\{Q_{i1}, \ldots, Q_{in}\}$), which can generate a new support set ($\{S'_{i1}, \ldots, S'_{in}\}$) and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

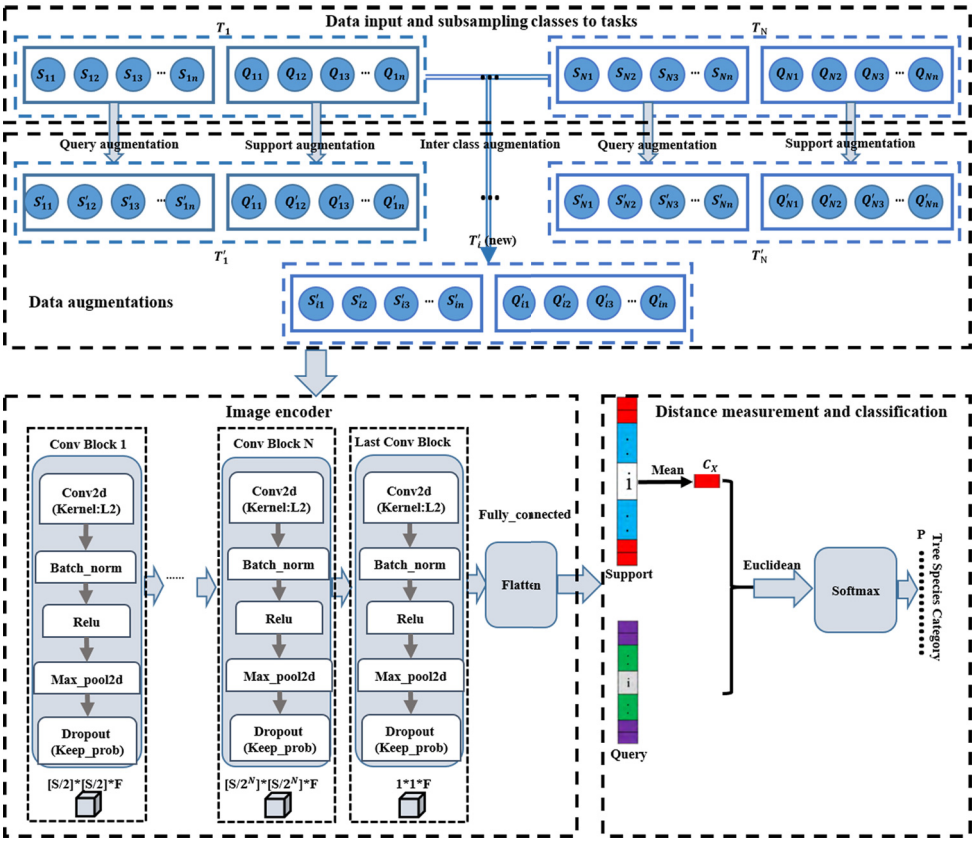



Fig. 2. PM overall architecture combing the P-Nets with data augmentation for forest tree species classification. It includes data input and subsampling classes to tasks, data augmentations, image encoder, distance measurement, and classification. Our work focuses on the data augmentations part, which is used to generate a new task set from the subsampled tasks in the previous part.

a query set [e.g., $T_1$ ($\{Q_{11}, \ldots, Q_{1n}\}$, $\{Q_{11}, \ldots, Q_{1n}\}$) to $T_1'$ ($\{Q'_{11}, \ldots, Q'_{1n}\}$, $\{Q'_{11}, \ldots, Q'_{1n}\}$)] through the inner-class augmentation mode. Simultaneously, in the data augmentation section, we get the new task [$T_i'$ (new)] by applying the inter-class augmentation mode to the task ($T_i$). The new task set ($\{T_1', \ldots, T_i'$ (new), $\ldots, T_N'\}$) generated in this way is used as data input for subsequently image encoder, distance measurement, and classification.

### B. Data Augmentation Techniques

As the deep learning network layers deepens, the parameters that need to be learned also increase, which is more likely to lead to overfitting. When the dataset is small, too many parameters will fit the features of the dataset rather than commonalities between the data. Data augmentation techniques are to enrich the diversity of samples in the dataset, to achieve the purpose of minimizing the issue of overfitting, although the application scenarios of each augmentation method are different. In this article, a series of traditional augmentation techniques (rotation, vertical flipping, and horizontal flipping) as well as SamplePairing, CutOut, MixUp, and CutMix are used to support and query augmentation mode, and techniques in inter-class augmentation mode, including SamplePairing, MixUp, and CutMix. It should be noted that we did not use SMOTE, because the samples of each tree species were balanced in this study. While Random Erasing and GridMask overlap with CutOut in theory, so this article takes CutOut as the representative to explore the problem of the data occlusion in tree species classification. Here, we introduce the augmentation techniques used in this article in detail.

*1) Rotation:* The training data are augmented by randomly rotating 90°, 180°, and 270° around the center of the images.

*2) Vertical Flipping and Horizontal Flipping:* The training data are augmented by flipping the images vertically and horizontally, respectively.

*3) SamplePairing:* The idea of SamplePairing is to superimpose two images randomly selected from the training set to synthesize a new sample by averaging the pixels, and using the label of the first image as the correct label of the synthesized image [21].

*4) CutOut:* The samples were augmented by randomly masking out square regions of the image to improve the localization ability of CNNs [22], which is similar to Random Erasing [23] and GridMask [24].

*5) MixUp:* A simple and data-independent augmentation method. In essence, MixUp constructs virtual samples on the convex combinations of paired samples and their labels to train neural networks. In this way, linear interpolation between training samples is encouraged [25]. In the P-Net, MixUp not only can be used between samples of the same category to ensure the generated new samples retain the original label, but also operate among different categories to produce a new category.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

*6) CutMix:* CutMix generates virtual samples by filling randomly cropped patches with other training samples. The label of the new sample is calculated proportionally based on the area of the patches from the two images [26]. Similar to MixUp, CutMix can be used for both inner-class and inter-class data augmentation, but the difference is that the images generated by CutMix are nearly natural (i.e., no zero-value padding and linear interpolation of image pairs).

### C. MaxUp for P-Net

MaxUp is a simple yet effective training strategy, which can promote the generalization of the model and alleviate the overfitting problem [14]. The idea of MaxUp is to find out the one with maximum loss from a series of augmented data and minimize its loss. Similar to adversarial training, MaxUp is also used to solve the issue of saddle point, where the loss is minimized relative to the parameters and maximized relative to the inputs [39]. In previous studies, the state-of-the-art classification performance of the ImageNet dataset was archived by combining MaxUp and CutMix, which is applied CutMix on one image for $m$ times, and selected the worst case to update the parameters of the model. Our study attempts to apply MaxUp to the data augmentation setting of the P-Net. In the input section, we select a series of better performance methods from the three data augmentation modes and their combinations as a data augmentation pool. Then, before training the model and updating the parameters, we apply augmentation methods to the training tasks to augment the randomly sampled support data, query data, and task diversity for $m$ times. Finally, we choose the data augmentation strategy with the maximum loss to augment each sample in the training set, and update the parameters according to the principle of minimizing the training loss. The solution of the minimax optimization issue is shown as follows:

$$\min_{\theta} \mathbb{E}_T = \left[ \max_{M \in A} L(\theta, M(Q)) \right] \tag{1}$$

where $\theta$ denotes the parameters, $L$ is the loss function, $M$ means an augmentation method in the pool $A$, $\mathbb{E}$ denotes the empirical risk, and $T$ represents a training task with support $S$ and query data $Q$, respectively.

Fig. 3 describes the loss calculation algorithm for a training task in P-Net with MaxUp comprehensively. Training set $D$ has $K$ categories and $N$ samples. With the random subsampling of $N_C$ classification tasks ($N_C \leq K$), the support set $S$ ($N_S$ samples for each category) and the query set $Q$ ($N_Q$ samples for each category) of each task are selected. We augment these inputs to generate new support set and query set, which may belong to the new task (inter-class augmentation) or not (support and query augmentation). The embedding function ($f_{\emptyset}$: $R^D \rightarrow R^Z$, $\emptyset$ is the learning parameters) is determined according to the image encoder (i.e., feature extraction framework based on different backbones) and the parameters of each layer. Without training, the corresponding loss of each data augmentation method is calculated according to the distance ($d$ represents the distance measurement function) between each class of prototype ($c_k \in R^Z$, where $Z$ represents the feature dimension of the last layer output) and its query set. At the same time, to prevent the randomness of the augmentation method itself, the loss calculation process is repeated $m$ times ($M_b^a$ denotes the $a$th augmentation using the $b$th method in the data augmentation pool $A$). Then, the loss is updated by the process of gradient descent, and the probability of the samples in the query set belonging to each category is obtained.

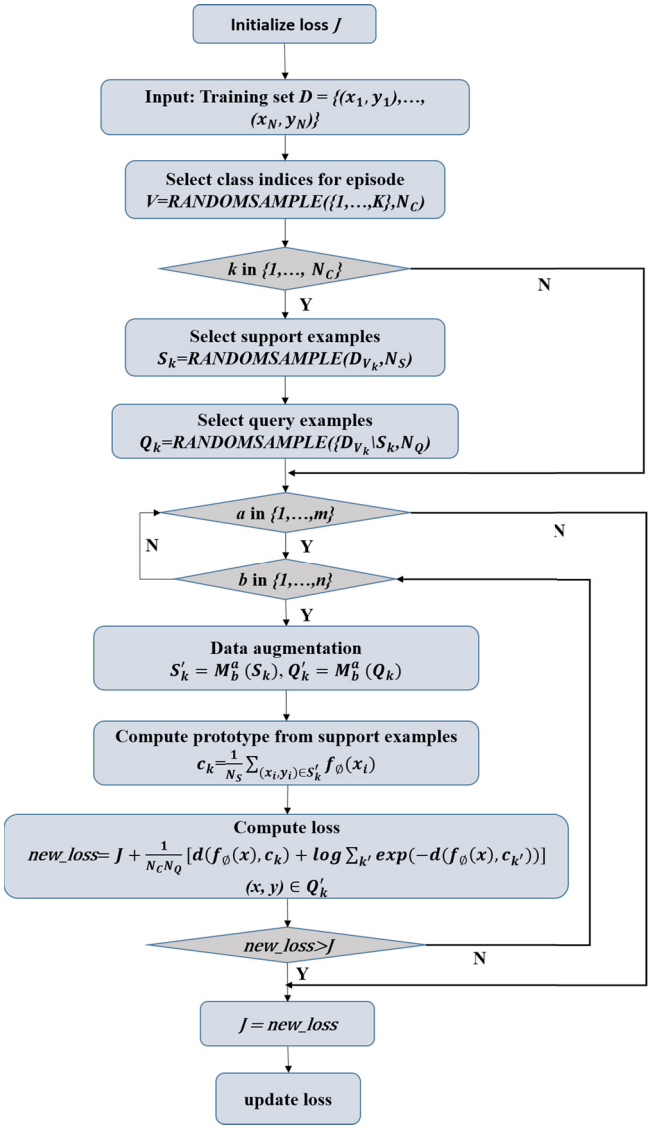



Fig. 3. Loss calculation algorithm for a training task in P-Nets with MaxUp.

## IV. Experimental Results and Discussion

### A. Experimental Design

Since this study focuses on data augmentation strategy and its application for tree species classification using airborne HSIs with the high spatial and spectral resolution. Given the limitations of spatial resolution, spectral resolution, and land cover types of the public hyperspectral datasets, we use GFF as the keynote dataset, and the IP, KSC, Botswana, and Chikusei datasets are only used to verify the proposed data augmentation strategy. Therefore, the dataset refers to GFF in the following descriptions unless special explanations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

TABLE V
EXPERIMENTS SCHEME

| Experiments | Name | Description |
|---|---|---|
| **A** | The impact of samples with different window sizes on classification performance | The window from 5×5 to 29×29, without any data augmentation, and applying the P-Net for classification with 5-shot, 10-shot, and 15-shot respectively. |
| **B** | Comparison of data augmentation modes | **$B_1$:** Comparison of augmentations in support augmentation mode.<br>**$B_2$:** Comparison of augmentations in query augmentation mode.<br>**$B_3$:** Comparison of augmentations in inter-class augmentation mode. |
| **C** | Combination of different data augmentation methods | Combining augmentations in order to understand the joint effect between them. |
| **D** | Using Proto-MaxUp to further improves classification performance | Evaluating our proposed Proto-MaxUp strategy by adjusting the hyper-parameter $m$ and selecting different data augmentation pools. |
| **E** | Proving the effectiveness of Proto-MaxUp strategy through other datasets | For the sake of verifying the effectiveness of our proposed data augmentation strategy sufficiently, we applied Proto-MaxUp to the other popular public hyperspectral datasets: Indian Pines, Kennedy Space Center, Botswana, and Chikusei. |
| **F** | Classification using Proto-MaxUp with different backbones | For the three different feature extraction backbones of the P-Net, we compared their baseline classification performance to augmentation with MaxUp. Finally, the classification maps corresponding to each model is constructed. |

The training settings of all models are consistent with the literature [3]. To quantitatively compare and evaluate the performance of each model, OA (2), Kappa coefficient [Kappa (3)], producer accuracy [PA (4)], and average accuracy [AA (5)] are used as testing accuracy measurements

$$\mathrm{OA} = \frac{\sum_{i=1}^{n} X_{ii}}{M} \tag{2}$$

$$\mathrm{Kappa} = \frac{M \sum_{i=1}^{n} X_{ii} - \sum_{i=1}^{n} X_{i+} \times X_{+i}}{M^2 - \sum_{i=1}^{n} X_{i+} \times X_{+i}} \tag{3}$$

$$\mathrm{PA} = \frac{X_{ii}}{X_{+i}} \tag{4}$$

$$\mathrm{AA} = \frac{\sum_{i=1}^{n} \mathrm{PA}}{n} \tag{5}$$

where $X_{ii}$, $X_{+i}$, and $X_{i+}$ denote the number of correctly classified samples, reference samples, and total samples (including correctly classified samples and misclassified samples) for the $i$th category ($i \in \{1, \ldots, n\}$, where $n$ is the number of categories) in the error matrix, respectively, and $M$ represents the totality of samples in the dataset.

The AA is not contained in the testing accuracy on the GFF dataset, because it has a balanced sample size for each class. Meanwhile, we take the AA of the last epoch (LEA) as the training accuracy. Thus, the gap between the LEA and the testing accuracy reflects the overfitting degree of the model. The detailed experiments scheme is shown in Table V.

### *B. Impact of Samples With Different Window Sizes on Classification Performance*

Spatial features and spectral information are the important characteristics of HSI and play a vital role in the classification tasks [43]–[46]. In the HSI classification task based on CNN, the expansion of the window enriches the spatial–spectral features of the samples, including effective features and extra noise that is not beneficial to the classification task. We usually need to increase the window size appropriately to ensure the classification performance of the P-Net; for instance, Tian *et al.* [3] obtained the best classification accuracy on a window size of 27 × 27 using a principle component analysis to reduce the spectral dimension of HSIs, while our newer research [2] performed parallel feature augmentation (attention mechanism) and data augmentations (rotation, vertical flipping, and horizontal flipping) on the original HSIs, which got the best performance on 17 × 17 window size. It achieved a balance among efficient feature extraction, rich spectral information, and fewer spatial features. However, when the training set and the testing set were reversed (20% for the training set and 80% for the testing set), the generalization of the model decreases significantly due to the changes in the distribution of the samples. Therefore, it is also necessary to determine the relationship between the spatial features of samples and the classification performance in the new training set while maintaining the spectral dimension invariant. In this study, the samples window size is from 5 × 5 to 29 × 29 without any data augmentation. Applying the P-Net for classification with 5-shot, 10-shot, and 15-shot, respectively, the results are shown in Fig. 4. In the 5-shot scenario, the testing accuracy OA reaches the peak in the 23 × 23 window. A similar upward trend occurred in the 10-shot and 15-shot tasks. The classification accuracy continued to increase until the window size of the samples is 23 × 23; subsequently, the classification accuracy decreases slightly from 23 × 23 to 29 × 29 window, because the invalid information in the larger window brings the noise to the classification task. It indicates that the spatial–spectral features from the 23 × 23 window are most conducive to the tree species classification in this dataset. Therefore, in the data augmentation experiments afterward,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

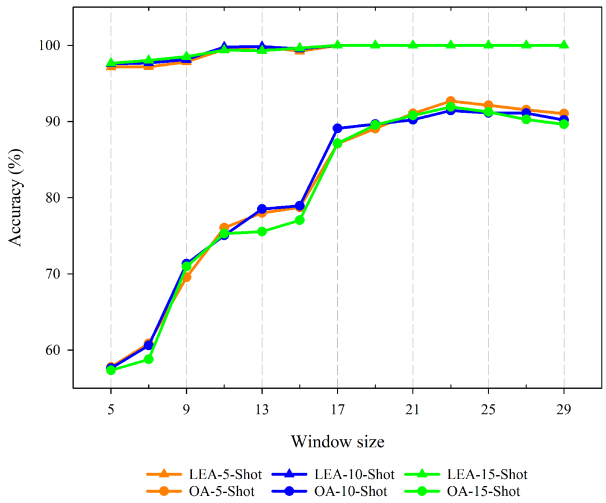



Fig. 4. Classification performance of P-Nets under different windows.

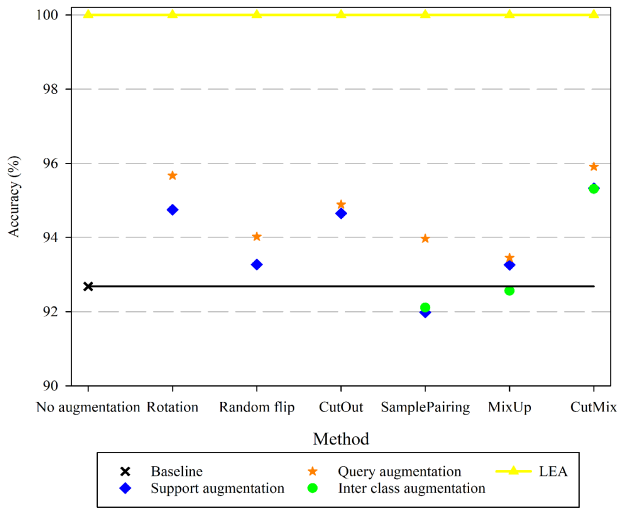



Fig. 5. Classification accuracy of data augmentation methods in each mode.

we adopt samples with 23 × 23 window size and the training tasks using 5-shot. Nevertheless, the issue of overfitting is apparent, because the testing accuracy is far lower than the training accuracy.

### C. Comparison of Data Augmentation Modes

Among the data augmentation methods mentioned earlier, SamplePairing, MixUp, and CutMix are methods to merge two images of different categories together. Therefore, they are more suitable for inter-class augmentation mode due to the image distribution having been changed tremendously caused by augmentation. On the other hand, the techniques of keeping the image distribution (e.g., rotation produces images that conform to the principle of invariance in CNN-based networks) are typically more suitable for inner-class augmentation mode. We empirically evaluated the performance of these techniques in support augmentation, query augmentation, and inter-class augmentation modes using a four-layer convolutional block networks backbone [47]. The results are shown in Fig. 5. The highest testing accuracy is obtained by query CutMix (OA of 95.91% and Kappa of 0.9550), which alleviates the overfitting of the model to a certain degree.

In the support augmentation modes, all methods except SamplePairing can obtain better classification accuracy than the baseline, and the optimal result is 95.33% of CutMix, while all methods including SamplePairing can achieve better performance in the query augmentation mode, in which the classification accuracy is generally higher than the support augmentation mode. We just adopt SamplePairing, MixUp, and CutMix to inter-class augmentation mode, because these methods can synthesize new samples from images with different categories. CutMix can also play a positive role in classification tasks with inter-class augmentation mode, but SampleParing and MixUp get opposite effects (the testing accuracy of inter-class augmentation mode using SampleParing and MixUp is lower than baseline). Therefore, the data augmentation methods of linear interpolation between categories are not helpful to the tree species classification using this dataset. It is worth noting that CutMix can obtain the best performance in all modes, which demonstrates that the method is more conducive to improve the generalization of the model by preserving the original data distribution in inner-class augmentation and generating samples that meet the new distribution in inter-class augmentation.

### D. Combining Augmentations

We combined augmentations to understand the joint effect between them after analyzing each data augmentation mode separately. The augmentation combinations are based on query CutMix (i.e., CutMix in query augmentation mode) and query rotation, because these methods are most prominently proved in Section IV-C. The classification accuracy results of query CutMix and query rotation combined with other effective augmentations are shown in Fig. 6. Interestingly, the query rotation combined with most methods can obtain better classification accuracy, while the query CutMix can only be effectively combined with query rotation and support CutMix. It demonstrates that the P-Net demands a careful data augmentation combination strategy. Simultaneously, the data augmentation combination is very important to prevent overfitting the classification model.

### E. Performance of PM Strategy

To further improve the classification accuracy, the MaxUp strategy is used in the P-Net, i.e., PM, to combine various data augmentation methods effectively and intelligently. The PM strategy is evaluated by adjusting the hyperparameter $m$ and selecting different data augmentation pools in the same experimental settings as mentioned earlier. We take NA as the baseline; "Single" means that each method in pool $A$ is of single augmentation, while "CutMix+" and "Rotation+" denote that pool $A$ consists of combination methods based on query CutMix and query rotation, respectively. As for the "Combined," pool $A$ is composed of the combinations of data augmentations that have better performance than the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
CLASSIFICATION PERFORMANCE OF AUGMENTATIONS COMBINATION USING PM STRATEGY

| Method | No augmentation | Single | CutMix+ | Rotation+ | Combined |
|---|---|---|---|---|---|
| *m* | - | 2 | 4 | 4 | 5 |
| LEA (%) | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 |
| OA (%) | 92.68 | 96.79 | 97.27 | 97.49 | **97.54** |
| Kappa | 0.9195 | 0.9647 | 0.9700 | 0.9724 | 0.9728 |
| *C. lanceolate* (%) | 89.86 | 96.48 | 98.20 | 97.82 | 98.96 |
| *P. elliottii* (%) | 92.02 | 94.14 | 95.78 | 97.26 | 96.88 |
| *P. massoniana* (%) | 97.24 | 99.94 | 100.00 | 100.00 | 100.00 |
| *E. urophylla* (%) | 79.66 | 93.86 | 95.82 | 93.10 | 97.04 |
| *E. grandis* (%) | 97.94 | 98.30 | 99.34 | 99.94 | 99.80 |
| *C. hystrix* (%) | 92.94 | 97.00 | 97.26 | 95.92 | 96.98 |
| *A. melanoxylon* (%) | 96.32 | 99.96 | 98.72 | 98.30 | 99.70 |
| *M. laosensis* (%) | 86.86 | 92.02 | 91.58 | 94.96 | 90.98 |
| Soft broadleaved (%) | 94.42 | 96.70 | 96.20 | 96.54 | 94.50 |
| Cutting-site (%) | 98.20 | 100.00 | 99.56 | 99.70 | 99.80 |
| Road (%) | 94.06 | 96.30 | 97.52 | 98.90 | 98.32 |

single augmentation in Section IV-D. The results are shown in Table VI, which indicates that the classification accuracy of all tree species is better than the baseline. Through adjusting $m$, we observe that the OA of the "Combined" is increased by nearly 5% compared with the baseline, which uses NA. Fig. 7 depicts the training and testing accuracy of the P-Net at each stage in our study, which proves the outstanding effect of the proposed data augmentation strategy on solving the issue of overfitting.

### *F. Visualizing Feature Maps With Different Data Augmentation Methods*

We explore the training benefits of these task-driven schemes by examining the distinguishability of feature extraction. To this point, we use t-distributed stochastic neighbor embedding (t-SNE) to reduce the dimension of the features from the last layer in the image encoder. The t-SNE method can not only preserve the local features of the dataset, but also map the data in high-dimensional space to low-dimensional space, in which the similar samples will be close to each other, while the samples with differences will be far away from each other. It can produce significantly better visualizations than the principal component analysis by reducing the aggregation of feature points in the center of the map. Fig. 8 shows the 2-D representation of the corresponding features using each data augmentation method, by which the distinguishability and the metric of discrimination between samples or categories can be visually displayed. It can be seen that the support augmentation [Fig. 8(b)] and query augmentation [Fig. 8(c)] modes improve the classification accuracy by narrowing the differences of samples from the same category, while the inter-class augmentation [Fig. 8(d)] mode is from the perspective of widening the gap between classes. It also reveals the slight advantages of augmentation combinations over the single one in the overall discrimination of the samples [comparison of Fig. 8(f) with Fig. 8(c), and Fig. 8(g) with Fig. 8(e)]. Fig. 8(h)–(k) illustrates that the discrimination of each tree species using the PM strategy is significantly better than the baseline [Fig. 8(a)]. The classification results with the pools of "single" [Fig. 8(h)] and "rotation+" [Fig. 8(j)] have a more obvious gap between categories than "CutMix+" [Fig. 8(i)] owing to the inter-class augmentation method is effective to expand the discrimination between classes. For overall samples discrimination, the pools of "CutMix+" and "Rotation+" are better than "Single," and it benefits from the advantages of augmentation combinations. The pool of "Combined" [Fig. 8(k)] makes full use of these advantages and shows a significant effect in narrowing the sample difference within the class and expanding the sample gap between the categories. Hence, such a data augmentation pool is the most favorable for the classification task in this research.

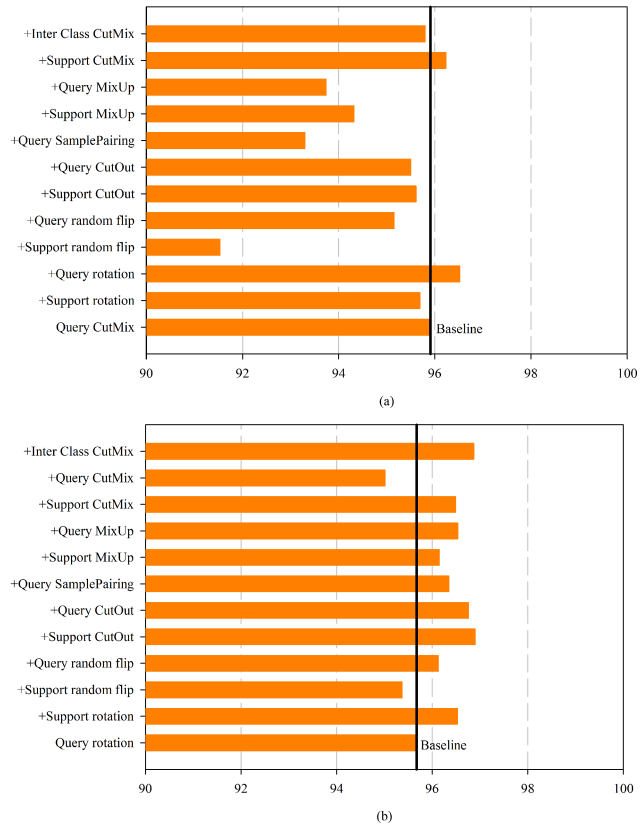



Fig. 6. Classification accuracy of augmentations combination based on (a) query CutMix and (b) query rotation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

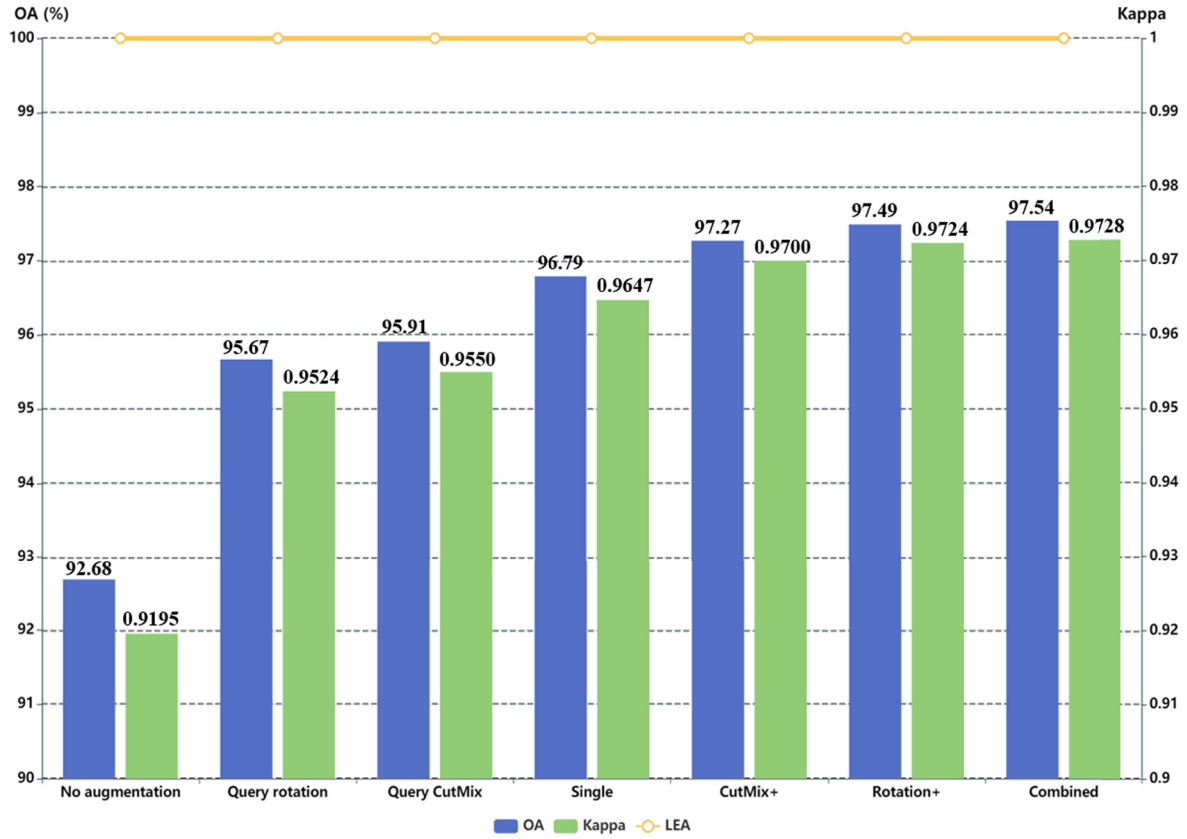



Fig. 7. Training and testing accuracy of the P-Nets at each stage.

TABLE VII
CLASSIFICATION PERFORMANCE OF IP, KSC, BOTSWANA, AND CHIKUSEI DATASET USING PM STRATEGY

| Dataset | Indian Pines | | Kennedy Space Center | | Botswana | | Chikusei | |
|---|---|---|---|---|---|---|---|---|
| Method | NA | PM | NA | PM | NA | PM | NA | PM |
| LEA (%) | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 |
| OA (%) | 94.12 | 98.18 | 89.78 | 98.95 | 87.90 | 97.54 | 98.08 | 99.67 |
| Kappa | 0.9326 | 0.9793 | 0.8861 | 0.9883 | 0.8687 | 0.9734 | 0.9799 | 0.9962 |
| AA (%) | 95.13 | 98.21 | 85.75 | 98.40 | 90.14 | 97.78 | 95.65 | 98.50 |
| Class 1 (%) | 85.71 | 90.32 | 99.17 | 99.45 | 95.36 | 96.39 | 96.56 | 97.58 |
| Class 2 (%) | 93.8 | 99.08 | 89.13 | 98.71 | 100.00 | 100.00 | 100.00 | 98.99 |
| Class 3 (%) | 86.87 | 98.49 | 74.77 | 99.18 | 97.98 | 100.00 | 61.01 | 82.51 |
| Class 4 (%) | 94.92 | 100.00 | 65.91 | 92.46 | 76.90 | 97.71 | 99.98 | 99.81 |
| Class 5 (%) | 95.81 | 98.26 | 79.03 | 96.72 | 94.58 | 98.83 | 98.87 | 99.86 |
| Class 6 (%) | 98.85 | 99.71 | 78.21 | 94.27 | 70.36 | 90.69 | 99.54 | 100.00 |
| Class 7 (%) | 100.00 | 100.00 | 73.33 | 100.00 | 99.61 | 100.00 | 100.00 | 100.00 |
| Class 8 (%) | 92.46 | 98.27 | 81.04 | 100.00 | 91.71 | 93.06 | 99.89 | 100.00 |
| Class 9 (%) | 100.00 | 100.00 | 89.98 | 100.00 | 79.64 | 100.00 | 99.98 | 99.93 |
| Class 10 (%) | 97.43 | 93.92 | 97.80 | 100.00 | 89.92 | 93.89 | 92.08 | 98.97 |
| Class 11 (%) | 91.78 | 97.84 | 98.03 | 99.25 | 80.60 | 100.00 | 95.31 | 100.00 |
| Class 12 (%) | 94.52 | 99.63 | 93.79 | 99.17 | 92.51 | 98.35 | 100.00 | 100.00 |
| Class 13 (%) | 100.00 | 99.49 | 94.62 | 100.00 | 92.81 | 100.00 | 99.91 | 99.59 |
| Class 14 (%) | 96.94 | 98.61 | - | - | 100.00 | 100.00 | 99.84 | 100.00 |
| Class 15 (%) | 100.00 | 100.00 | - | - | - | - | 100.00 | 100.00 |
| Class 16 (%) | 92.96 | 97.75 | - | - | - | - | 100.00 | 100.00 |
| Class 17 (%) | - | - | - | - | - | - | 94.70 | 96.80 |
| Class 18 (%) | - | - | - | - | - | - | 86.59 | 97.39 |
| Class 19 (%) | - | - | - | - | - | - | 93.18 | 100.00 |

### *G. Verifying the Effectiveness of PM Strategy Through Other Datasets*

IP, KSC, Botswana, and Chikusei, four public datasets of the HSI, are widely used in the research of deep learning classification methods. In this article, we further verify the effectiveness of our proposed data augmentation strategy on these cross-domain datasets. Data augmentation methods may be very effective on specific datasets, but weak in other tasks due to differences in data distribution [31], [39], [48]. Thus, testing on different datasets is crucial. To this point, we apply the abovementioned experimental settings to IP,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

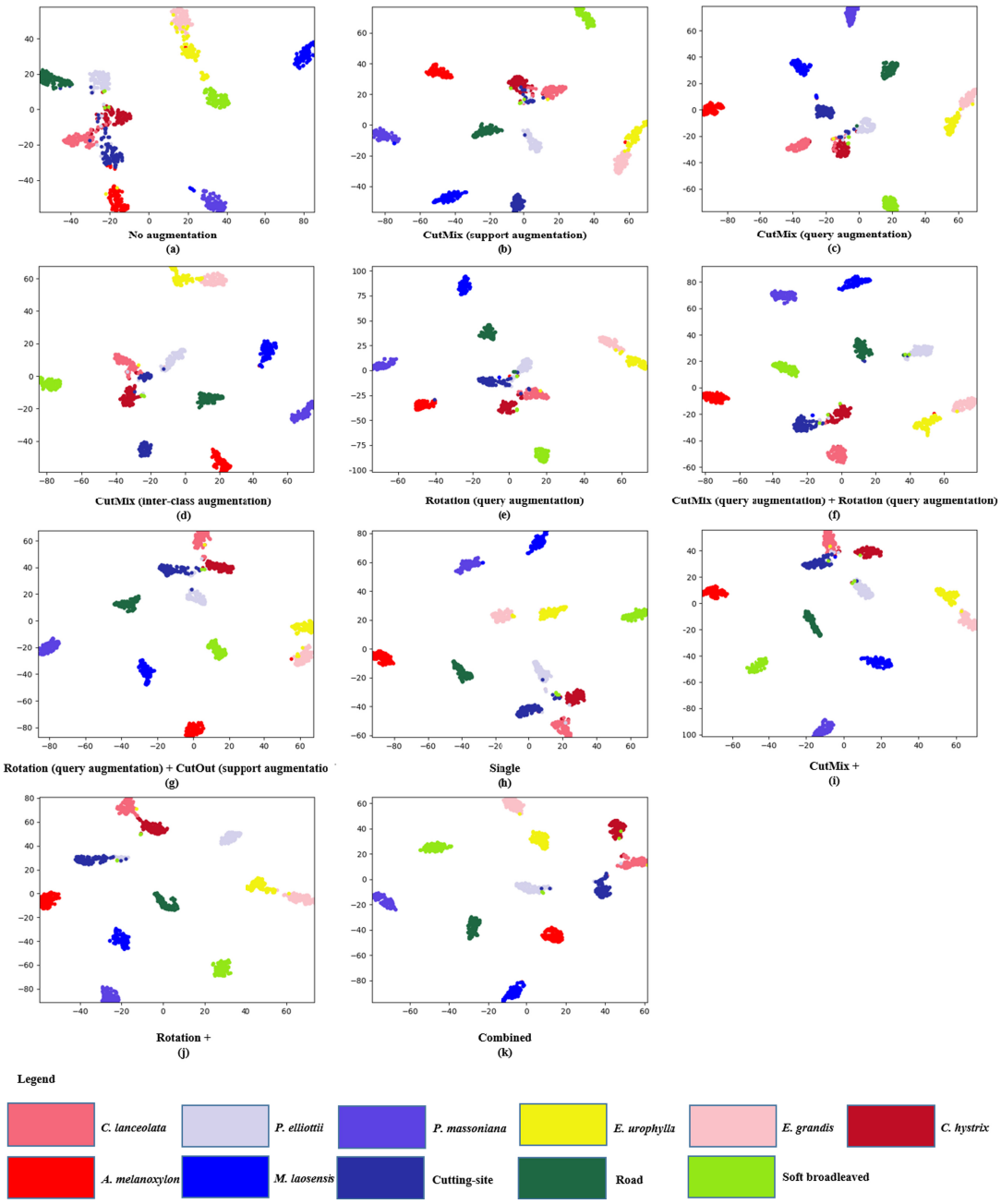



Fig. 8. Visualizing the feature map of the last layer in P-Nets using t-SNE. The $x$-axis represents the first dimensional feature, and the $y$-axis is the second dimensional feature. (a) NA. (b) CutMix in support augmentation mode. (c) CutMix in query augmentation mode. (d) CutMix in inter-class augmentation mode. (e) Rotation in query augmentation mode. (f) CutMix combined with rotation in query augmentation mode. (g) Rotation in query augmentation mode combined with CutOut in support augmentation mode. (h) PM with a pool of single augmentation. (i) PM with a pool of combination methods based on CutMix in query augmentation mode. (j) PM with a pool of combination methods based on rotation in query augmentation mode. (k) PM with a pool of combination methods that have better performance than the single augmentation.

KSC, Botswana, and Chikusei in a similar way to obtain the corresponding classification accuracy, as shown in Table VII (NA means no augmentation, and PM is Proto-MaxUp). What needs to be added is that the size of each sample in the IP, KSC, Botswana, and Chikusei is 11 $\times$ 11 following the kinds of the literature [49], [50]. We can find out that our proposed data augmentation strategy can greatly improve the testing accuracy on all datasets. The magnitude of this increase varies according to the distribution of data. Focusing on the IP scene, we can observe that the testing accuracies of OA and AA have increased by more than 4%. In the KSC scenario, the OA and AA even increased by more than 9% and 12%, respectively. As for Botswana, the dataset carried by satellite, we can obtain the OA and AA increased by more than 9% and 7%. Applying our proposed strategy to the Chikusei dataset, the OA and AA were improved to 99.67%

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

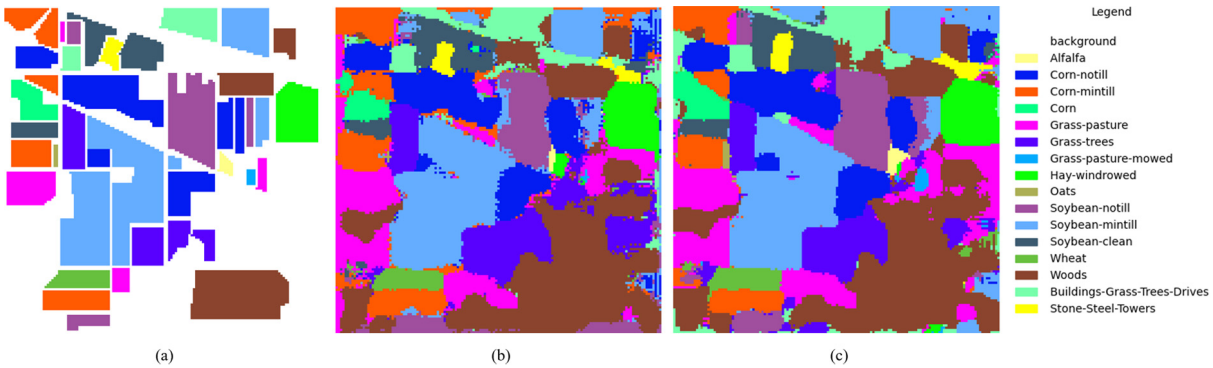



Fig. 9. Classification maps of P-Nets on IP dataset. (a) Ground truth. (b) P-Nets with NA. (c) P-Nets with PM.

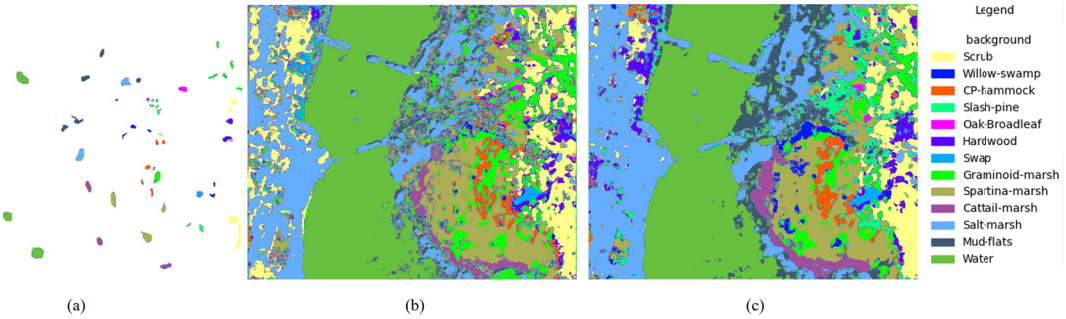



Fig. 10. Classification maps of P-Nets on KSC dataset. (a) Ground truth. (b) P-Nets with NA. (c) P-Nets with PM.

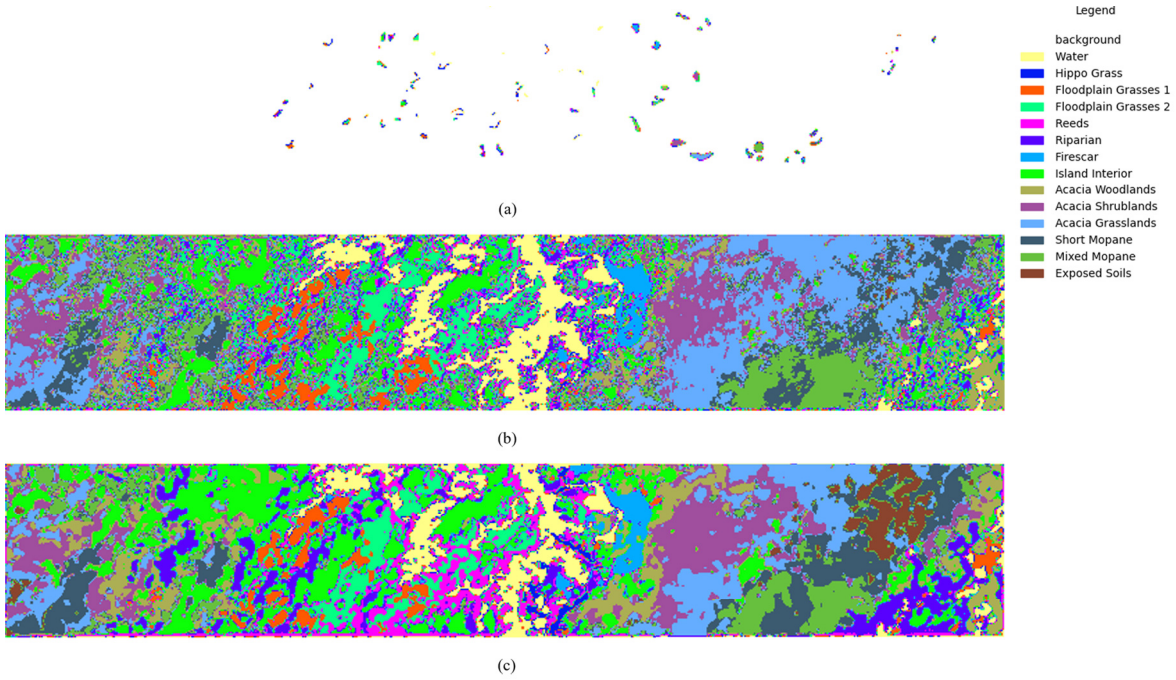



Fig. 11. Classification maps of P-Nets on Botswana dataset. (a) Ground truth. (b) P-Nets with NA. (c) P-Nets with PM.

and 98.5%, respectively, although the improvement is not as great as other datasets. This is enough to illustrate the effectiveness of our proposed PM on the public hyperspectral datasets.

Furthermore, to compare the classification performance visually, we perform pixel-by-pixel prediction on IP, KSC, Botswana, and Chikusei images to generate classification maps, respectively, as shown in Figs. 9–12. It is worth noting that the zero-valued pixels of the image edge in the Chikusei dataset are classified as man-made (blue) or man-made (dark) due to the cluster center of the model being shifted by the data augmentation strategy. In the real life, this difference should be ignored. As this article focuses on the task of tree species classification based on GFF, we will not discuss the classification results of other datasets for method verification in detail.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

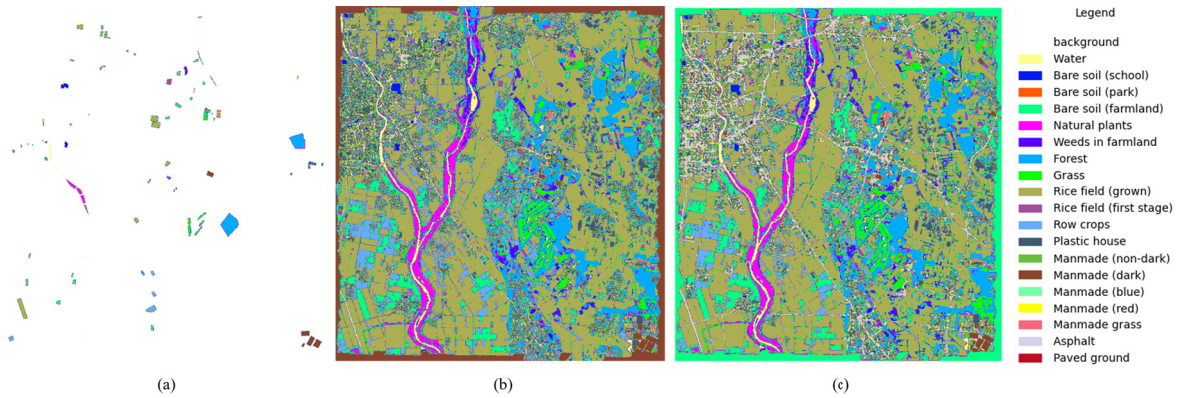



Fig. 12. Classification maps of P-Nets on Chikusei dataset. (a) Ground truth. (b) P-Nets with NA. (c) P-Nets with PM.

TABLE VIII
CLASSIFICATION PERFORMANCE USING PM WITH DIFFERENT BACKBONES

| Method | CNN-4 | | 64-64-64-64 | | CBAM+64-64-64-64 | |
|---|---|---|---|---|---|---|
| | NA | PM | NA | PM | NA | PM |
| LEA (%) | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 |
| OA (%) | 92.69 | 97.62 | 92.68 | 97.54 | 93.77 | **98.08** |
| Kappa | 0.9192 | 0.9738 | 0.9195 | 0.9728 | 0.9315 | 0.9789 |
| *C. lanceolate* (%) | 85.68 | 94.56 | 89.86 | **98.96** | 92.88 | 98.76 |
| *P. elliottii* (%) | 84.98 | **99.82** | 92.02 | 96.88 | 90.10 | 98.92 |
| *P. massoniana* (%) | 98.46 | 99.68 | 97.24 | **100.00** | 96.90 | 99.14 |
| *E. urophylla* (%) | 88.40 | 96.00 | 79.66 | 97.04 | 86.00 | **97.58** |
| *E. grandis* (%) | 98.42 | **100.00** | 97.94 | 99.80 | 98.34 | 99.98 |
| *C. hystrix* (%) | 92.02 | 95.74 | 92.94 | **96.98** | 92.60 | 96.16 |
| *A. melanoxylon* (%) | 96.82 | **99.82** | 96.32 | 99.70 | 94.36 | 99.58 |
| *M. laosensis* (%) | 86.94 | **93.12** | 86.86 | 90.98 | 89.98 | 92.56 |
| Soft broadleaved (%) | 95.18 | **98.98** | 94.42 | 94.50 | 95.58 | 98.86 |
| Cutting-site (%) | 98.96 | 100.00 | 98.20 | 99.80 | 99.38 | 100.00 |
| Road (%) | 93.76 | 96.08 | 94.06 | 98.32 | 95.36 | 97.32 |

### *H. Classification Using PM With Different Backbones*

In solving the issue of overfitting, effective feature extraction is as important as data augmentation. We use the proposed data augmentation strategy to improve the performance of three P-Nets based on different backbones.

*1) CNN-4:* Bertinetto *et al.* [32] proposed a four-layer CNN, in which each layer has 96, 192, 384, and 512 filters, respectively.

*2) 64-64-64-64:* Snell *et al.* [47] proposed a four-layer CNN backbone, in which each convolutional layer has 64 filters.

*3) CBAM+64-64-64-64:* The convolutional block attention module (CBAM) was embedded into the convolutional blocks [2].

On the GFF dataset, we obtain the classification performance of their baseline (NA) and augmentation with MaxUp (PM), and the results are shown in Table VIII. Without using PM, the method with CBAM+64-64-64-64 as the backbone can also achieve better performance over others. This shows that the efficient feature extraction technology is conducive to classification tasks. In all cases with PM, the performance of existing methods is improved by more than 4%, even close to 5%. In the model with the backbone of CBAM+64-64-64-64, the classification accuracy reaches the highest OA of 98.08% and the Kappa of 0.9789. In particular, the proposed data augmentation strategy can achieve higher classification accuracy than the baseline on each tree species. It demonstrates that the data augmentation strategy proposed in this study can be combined with feature augmentation techniques to train accurate classification models. However, if we compare the classification performance with different backbones, we will find that the classification accuracy gap of each tree species is fluctuating. The performance improvement obtained by the efficient feature extraction method is only reflected in the OA rather than each tree species, because each training task that uses all categories results in the loss calculation and parameters updating often corresponding to the overall feature representation.

The classification maps are constructed by predicting the label of airborne HSI pixel by pixel until that covers the entire study area. As shown in Fig. 13, referring to the data of subcompartment surveys, we can see that the models with PM are significantly better than the corresponding baseline models in classification effect. Without data augmentation, the boundary between the coniferous species *C. lanceolate* and *P. elliottii* is unclear, and a large number of *P. elliottii* is misclassified. As for broadleaf species, *E. urophylla* and *E. grandis* are mixed. At the same time, the confusion of coniferous and broadleaf species is also existing, such as *P. Massoniana* with *M. laosensis* and *A. melanoxylon* with *E. grandis*. When using

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

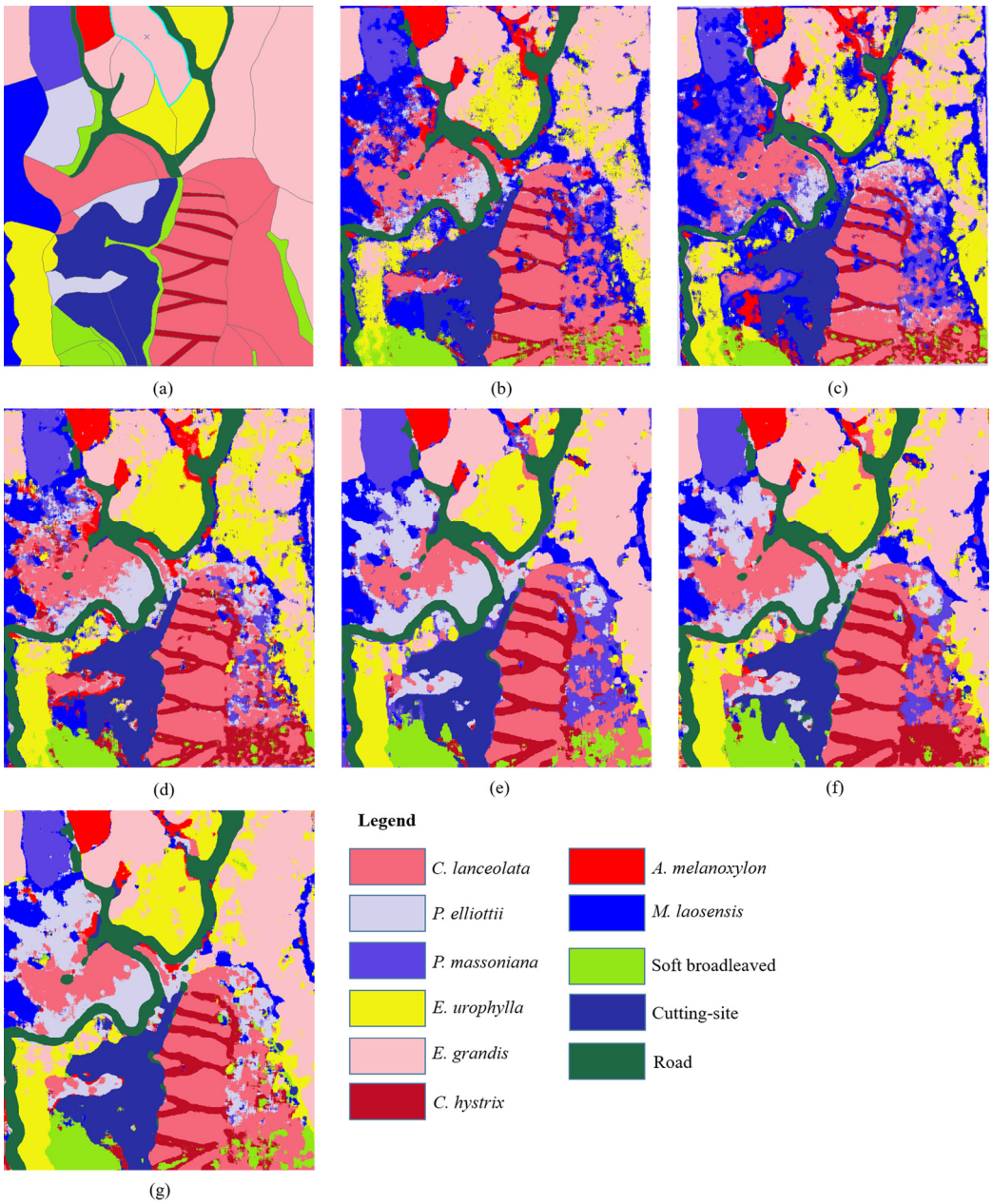



Fig. 13. Classification maps of P-Nets with different backbones. (a) Forest sub-compartment surveys data. (b) CNN-4 with NA. (c) 64-64-64-64 with NA. (d) CBAM+64-64-64-64 with NA. (e) CNN-4 with PM. (f) 64-64-64-64 with PM. (g) CBAM+64-64-64-64 with PM.

PM for data augmentation, the model obtained with CNN-4 as the backbone has a better classification effect on broadleaf species than 64-64-64-64, which is more distinguishable on coniferous species. CBAM+64-64-64-64 has the overall classification performance advantage, which is manifested in less noise and island patches.

## V. Conclusion

In this work, we minimized the issue of overfitting in the classification task with P-Net from the perspective of data augmentation. First, we decomposed the data input of each part in the P-Net and determined three data augmentation modes, i.e., support, query, and inter-class augmentation. Then, several data augmentation pools (e.g., "single," "CutMix+," "Rotation+," and "combined") were established by selecting the data augmentation methods or their combinations that perform better than baseline in each mode, respectively. We proposed a PM strategy to combine the methods in these pools and obtained further performance improvement subsequently. To verify the robustness and validity of our proposed data augmentation strategy sufficiently, we applied PM to the other four popular public hyperspectral datasets, such as IP, KSC, Botswana, and Chikusei, and achieved excellent results. Finally, we applied this data augmentation strategy to the training tasks of P-Net with different backbones, the best testing accuracy OA is 98.08%, and the Kappa is 0.9789 using CBAM+64-64-64-64 backbone. The difference between training accuracy and test accuracy is less than 2%. It demonstrates that the proposed framework of the PM data argumentation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

method and CBAM+64-64-64-64 network backbone significantly improved the classification performance of the P-Net.

## ACKNOWLEDGMENT

The authors would like to thank Lin Zhao, Zhengqi Guo, Xuemei Zhou, and Linghan Gao from Beijing Forestry University, Beijing, China, and Dr. Lei Zhao from the Institute of Forest Resource Information Techniques, Chinese Academy of Forestry, Beijing, for their assistance on data collection for the Gaofeng Forest Farm dataset. They would also like to thank the image providers and the samples producers of the Indian Pines, Kennedy Space Center, Botswana, and Chikusei datasets (available online from http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes and https://naotoyokoya.com/Download.html).

## REFERENCES

[1] B. Zhang, L. Zhao, and X. Zhang, "Three-dimensional convolutional neural network model for tree species classification using airborne hyperspectral images," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111938, doi: 10.1016/j.rse.2020.111938.

[2] L. Chen, X. Tian, G. Chai, X. Zhang, and E. Chen, "A new CBAM-P-Net model for few-shot forest species classification using airborne hyperspectral images," *Remote Sens.*, vol. 13, no. 7, p. 1269, Mar. 2021. [Online]. Available: https://www.mdpi.com/2072-4292/13/7/1269

[3] X. Tian, L. Chen, X. Zhang, and E. Chen, "Improved prototypical network model for forest species classification in complex stand," *Remote Sens.*, vol. 12, no. 22, p. 3839, Nov. 2020, doi: 10.3390/rs12223839.

[4] J. Rajendran, A. Irpan, and E. Jang, "Meta-learning requires meta-augmentation," 2020, *arXiv:2007.05549*.

[5] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, pp. 1929–1958, Jun. 2014.

[6] L. Wan, M. Zeiler, S. Zhang, Y. Le Cun, and R. Fergus, "Regularization of neural networks using dropconnect," in *Proc. Int. Conf. Mach. Learn.*, 2013, pp. 1058–1066.

[7] I. Goodfellow, D. Warde-Farley, M. Mirza, A. Courville, and Y. Bengio, "Maxout networks," in *Proc. 30th Int. Conf. Mach. Learn.*, 2013, pp. 1319–1327.

[8] G. Huang, Y. Sun, Z. Liu, D. Sedra, and K. Q. Weinberger, "Deep networks with stochastic depth," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 646–661.

[9] G. Larsson, M. Maire, and G. Shakhnarovich, "FractalNet: Ultra-deep neural networks without residuals," 2016, *arXiv:1605.07648*.

[10] B. Zoph, V. Vasudevan, J. Shlens, and Q. V. Le, "Learning transferable architectures for scalable image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8697–8710.

[11] X. Gastaldi, "Shake-shake regularization," 2017, *arXiv:1705.07485*.

[12] Y. Yamada, M. Iwamura, and K. Kise, "Shakedrop regularization," in *Proc. 6th Int. Conf. Learn. Represent. (ICLR)-Workshop Track*, Vancouver, BC, Canada, 2018.

[13] G. Ghiasi, T.-Y. Lin, and Q. V. Le, "DropBlock: A regularization method for convolutional networks," 2018, *arXiv:1810.12890*.

[14] C. Gong, T. Ren, M. Ye, and Q. Liu, "MaxUp: A simple way to improve generalization of neural network training," 2020, *arXiv:2002.09024*.

[15] M. Zahangir Alom *et al.*, "The history began from AlexNet: A comprehensive survey on deep learning approaches," 2018, *arXiv:1803.01164*.

[16] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[17] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 1–9.

[18] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[19] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 4700–4708.

[20] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "Smote: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, Dec. 2002.

[21] H. Inoue, "Data augmentation by pairing samples for images classification," 2018, *arXiv:1801.02929*.

[22] T. DeVries and G. W. Taylor, "Improved regularization of convolutional neural networks with cutout," 2017, *arXiv:1708.04552*.

[23] Z. Zhong, L. Zheng, G. Kang, S. Li, and Y. Yang, "Random erasing data augmentation," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 13001–13008.

[24] P. Chen, S. Liu, H. Zhao, and J. Jia, "GridMask data augmentation," 2020, *arXiv:2001.04086*.

[25] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "Mixup: Beyond empirical risk minimization," 2017, *arXiv:1710.09412*.

[26] S. Yun, D. Han, S. Chun, S. J. Oh, Y. Yoo, and J. Choe, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6023–6032.

[27] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016, doi: 10.1109/TGRS.2016.2584107.

[28] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017, doi: 10.1109/TGRS.2016.2616355.

[29] J. Acquarelli, E. Marchiori, L. M. C. Buydens, T. Tran, and T. Van Laarhoven, "Convolutional neural networks and data augmentation for spectral-spatial classification of hyperspectral images," *Networks*, vol. 16, pp. 21–40, 2017.

[30] J. Nalepa, M. Myller, and M. Kawulok, "Training- and test-time data augmentation for hyperspectral image segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 292–296, Feb. 2020, doi: 10.1109/LGRS.2019.2921011.

[31] W. Wang, X. Liu, and X. Mou, "Data augmentation and spectral structure features for limited samples hyperspectral classification," *Remote Sens.*, vol. 13, no. 4, p. 547, Feb. 2021, doi: 10.3390/rs13040547.

[32] L. Bertinetto, J. F. Henriques, P. H. S. Torr, and A. Vedaldi, "Meta-learning with differentiable closed-form solvers," 2018, *arXiv:1805.08136*.

[33] K. Lee, S. Maji, A. Ravichandran, and S. Soatto, "Meta-learning with differentiable convex optimization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 10657–10665.

[34] H. Tang, Y. Li, X. Han, Q. Huang, and W. Xie, "A spatial–spectral prototypical network for hyperspectral remote sensing image," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 1, pp. 167–171, Jan. 2020, doi: 10.1109/LGRS.2019.2916083.

[35] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019, doi: 10.1109/TGRS.2018.2872830.

[36] S. Liu, Q. Shi, and L. Zhang, "Few-shot hyperspectral image classification with unknown classes using multitask deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5085–5102, Jun. 2021, doi: 10.1109/TGRS.2020.3018879.

[37] T. Gao, X. Han, Z. Liu, and M. Sun, "Hybrid attention-based prototypical networks for noisy few-shot relation classification," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, 2019, pp. 6407–6414, doi: 10.1609/aaai.v33i01.33016407.

[38] G. Song, Z. Tao, X. Huang, G. Cao, W. Liu, and L. Yang, "Hybrid attention-based prototypical network for unfamiliar restaurant food image few-shot recognition," *IEEE Access*, vol. 8, pp. 14893–14900, 2020, doi: 10.1109/ACCESS.2020.2964836.

[39] R. Ni, M. Goldblum, A. Sharaf, K. Kong, and T. Goldstein, "Data augmentation for meta-learning," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8152–8161.

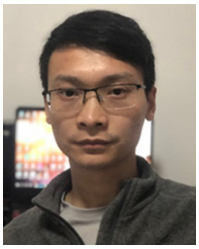
[40] V. H. Ayma, V. A. Ayma, and J. Gutierrez, "Dimensionality reduction via an orthogonal autoencoder approach for hyperspectral image classification," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 43, pp. 1–6, Aug. 2020, doi: 10.5194/isprs-archives-XLIII-B3-2020-357-2020.

[41] N. Kakhani and M. Mokhtarzade, "A new neuro-fuzzy-based classification approach for hyperspectral remote sensing images," *J. Earth Syst. Sci.*, vol. 128, no. 2, pp. 1–21, Mar. 2019, doi: 10.1007/s12040-018-1054-9.

[42] H. Hasan, H. Z. M. Shafri, and M. Habshi, "A comparison between support vector machine (SVM) and convolutional neural network (CNN) models for hyperspectral image classification," in *Proc. IOP Conf., Earth Environ. Sci.*, vol. 357, 2019, Art. no. 012035, doi: 10.1088/1755-1315/357/1/012035.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.

[43] X. Ma, A. Fu, J. Wang, H. Wang, and B. Yin, "Hyperspectral image classification based on deep deconvolution network with skip architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4781–4791, Aug. 2018.
[44] S. Mei, J. Ji, J. Hou, X. Li, and Q. Du, "Learning sensor-specific spatial–spectral features of hyperspectral images via convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4520–4533, Aug. 2017, doi: 10.1109/TGRS.2017.2693346.
[45] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. J. Plaza, and F. Pla, "Deep pyramidal residual networks for spectral-spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Feb. 2019.
[46] L. Mou and X. X. Zhu, "Learning to pay attention on spectral domain: A spectral attention module-based convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 110–122, Sep. 2020.
[47] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. 31st Conf. Neural Inf. Process. Syst.*, Long Beach, CA, USA, 2017, pp. 4077–4087.
[48] J. Nalepa, M. Myller, M. Kawulok, and B. Smolka, "On data augmentation for segmenting hyperspectral images," *Proc. SPIE*, vol. 10996, May 2019, Art. no. 1099609, doi: 10.1117/12.2519517.
[49] M. E. Paoletti, J. M. Haut, X. Tao, J. Plaza, and A. Plaza, "FLOP-reduction through memory allocations within CNN for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5938–5952, Jul. 2021, doi: 10.1109/TGRS.2020.3024730.
[50] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021, doi: 10.1109/TGRS.2020.3015157.



**Long Chen** received the B.S. degree in geographic information system from the Central South University of Forestry and Technology, Changsha, China, in 2016. He is currently pursuing the master's degree with Beijing Forestry University, Beijing, China.

His research interests include computer vision, graphic image processing, deep learning, and web development.



**Yuxin Wei** was born in Dezhou, Shandong, China, in 1998. She received the B.S. degree in geomatics engineering from the Hebei University of Technology, Tianjin, China, in 2020. She is currently pursuing the master's degree in agricultural engineering and information technology with Beijing Forestry University, Beijing, China.

Her research interests include deep learning, image classification, and data fusion.



**Zongqi Yao** received the B.S. degree in surveying and mapping engineering from the China University of Mining and Technology-Beijing, Beijing, China, in 2020. He is currently pursuing the M.S. degree in cartography and geographic information systems with Beijing Forestry University, Beijing.

His research interests include remote sensing monitoring of forest change.



**Erxue Chen** (Member, IEEE) received the B.S. and M.Agr. degrees in forest management and statistics forestry from Shandong Agricultural University, Taian, China, in 1992 and 1995, respectively, and the Ph.D. degree in synthetic aperture radar (SAR) image geocoding and topography radiometric correction algorithm from the Chinese Academy of Forestry, Beijing, China, in 2004.

He is currently a Professor with the Institute of Forest Resource Information Techniques, Chinese Academy of Forestry. He has taught courses including radar remote sensing of forest resources and radar remote sensing of vegetation. His research interests include polarimetric SAR interferometry (PolInSAR) and SAR data processing, and forest applications.



**Xiaoli Zhang** (Member, IEEE) received the B.S. degree in computer science from Hebei University, Baoding, China, in 1989, and the Ph.D. degree in forest management science from Beijing Forestry University, Beijing, China, in 1998.

From 1998 to 1999, she was a Post-Doctoral Research Fellow with the Institute of Remote Sensing Application, Chinese Academy of Science (CAS), Beijing, China. From 2006 to 2007, she was a Visiting Scholar in remote sensing and intelligent analysis with Lincoln University, Lincoln, New Zealand. She is currently a Full Professor with Beijing Forestry University. Her research interests include forest resources and ecological environment remote sensing monitoring and spatial prediction, forest disease and pest remote sensing monitoring and risk assessment, and vegetation quantitative remote sensing.

Dr. Zhang is a Councilor of the Associate on Environment Remote Sensing of China (AERSC), a Supervisor of the Beijing Society for Information Technology in Agriculture, and a member of the Forest Management Branch of Chinese Society of Forestry. She was a recipient of the National Scientific and Technological Progress Award, the Ministry of Education (MOE) Scientific and Technological Progress Award, the Beijing Science and Technology Award, the Forest Science and Technology Award for Chinese Youth, and the MOE New Century Excellent Talents (NCET).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:48:32 UTC from IEEE Xplore. Restrictions apply.